

S475/1

SUB MATH

PAPER 1

2HOURS 40 MINUTES



Apex Examination Council

PRE-MOCK EXAMINATION 2026 SET 2

UGANDA ADVANCED CERTIFICATE OF EDUCATION

SUBSIDIARY MATHEMATICS

TIME: 2 HOURS 15 MINUTES

INSTRUCTIONS TO CANDIDATES

- This paper consists of four sections: A, B, C and D.
- The paper contains six examination items.
- Section A has two items. Answer ONE item only.
- Section B has two items. Answer ONE item only.
- Section C has one compulsory item.
- Section D has one compulsory item.
- Answer FOUR items in all.
- Any additional item(s) answered will not be scored.
- Silent non-programmable scientific calculators and mathematical tables may be used.

SECTION A

ANSWER ONE ITEM ONLY

ITEM 1

A supermarket chain operating branches in Kampala, Jinja, Mbarara, Gulu and Mbale is monitoring monthly sales of cooking oil. Management wants to analyse sales performance and forecast future demand.

The monthly sales (in cartons) for ten consecutive months were recorded as follows:

120, 135, 140, 150, 165, 175, 180, 195, 210, 225

The company also found that sales revenue Y (million shillings) and advertising expenditure X (million shillings) are related as shown below:

X	2	4	6	8	10
Y	20	28	35	45	52

Tasks

- (a) Determine the mean and standard deviation of the monthly sales.
- (b) Determine the equation of the regression line of Y on X.
- (c) Estimate the sales revenue when advertising expenditure is 12 million shillings.
- (d) Comment on the relationship between advertising expenditure and sales revenue.

ITEM 2

A health organisation is studying the occurrence of malaria in a district. Records show that the probability that a household experiences malaria in a given month is 0.25.

A sample of 12 households is selected.

Tasks

- (a) Determine the probability that exactly four households experience malaria.
- (b) Determine the probability that at least three households experience malaria.
- (c) Calculate the expected number of households that experience malaria.
- (d) Determine the variance of the distribution.

SECTION B

ANSWER ONE ITEM ONLY

ITEM 3

A teachers' savings association intends to accumulate UGX 36,000,000 in 12 months.

Members agree to contribute equal monthly amounts that increase by a constant value every month. The first contribution is x shillings and the monthly increase is d shillings.

The sixth contribution is UGX 120,000 and the twelfth contribution is UGX 240,000.

Tasks

- (a) Determine the values of x and d .
- (b) Determine the total contribution made by one member during the twelve months.
- (c) If there are 25 members, determine whether the target amount will be achieved.
- (d) Calculate the shortfall or surplus.

ITEM 4

A company produces bottled fruit juice. Market research shows that weekly profit P (in millions of shillings) is given by

$$P(x) = 60 + 24x - x^2$$

where x is the amount invested in advertising.

The company also estimates that customer growth follows

$$N = 500(1.08)^t$$

where t is measured in months.

Tasks

- (a) Determine the value of x that gives maximum profit.
- (b) Determine the maximum profit.
- (c) Determine the number of customers after 18 months.
- (d) Determine the time required for the customer base to double.

SECTION C
COMPULSORY

ITEM 5

A landscape architect is designing a recreational park. The boundary of part of the park is represented by the curve

$$y = x^2 - 6x + 8$$

and the x-axis.

The architect also studies the growth of ornamental trees. The rate of growth of the height h metres is proportional to the difference between the maximum height of 12 metres and the current height.

Initially, the trees are 2 metres high. After 3 years, the average height is 5 metres.

Tasks

- (a) Determine the area enclosed between the curve and the x-axis.
- (b) Form a differential equation representing the growth process.
- (c) Determine the growth model.
- (d) Estimate the height of the trees after 6 years.

SECTION D
COMPULSORY

ITEM 6

Two surveyors A and B leave the same point O.

Surveyor A travels 8 km in the direction of vector

$$2\mathbf{i} + 3\mathbf{j}$$

Surveyor B travels 10 km in the direction of vector

$$5\mathbf{i} + 12\mathbf{j}$$

After reaching their respective positions, they communicate by radio.

Later, Surveyor A observes the top of a communication mast. From one point, the angle of elevation is 32° .

After moving 40 metres closer to the mast along the same straight line, the angle of elevation becomes 56° .

Tasks

- (a) Determine the distance between the two surveyors.
- (b) Determine the height of the communication mast.
- (c) Determine the distance of the first observation point from the base of the mast.
- (d) Comment on the usefulness of vectors and trigonometry in surveying and navigation.

END